

CHAMPAIGN 9 SW, IL, USA

WMO: 724360

Lat: 40.0528N

Lon: 88.3729W

Elev: 700

StdP: 14.33

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 54808

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.0	(c) 3.7	(d) -8.0	(e) 3.7	(f) -2.4	(g) -3.0	(h) 4.8	(i) 3.5	(j) 24.1	(k) 25.2	(l) 21.8	(m) 22.6	(n) 5.3	(o) N/A	(p) 0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.8	(c) 90.3	(d) 75.6	(e) 88.1	(f) 75.0	(g) 86.0	(h) 73.8	(i) 79.3	(j) 87.6	(k) 77.5	(l) 85.4	(m) 75.9	(n) 83.3	(o) 7.0	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	144.4	85.5	75.2	135.6	82.9	73.5	127.9	80.9	43.3	87.5	41.5	85.4	39.8	83.5	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	17.9	15.6	DB	-10.6	92.9	6.7	2.7	-15.4	94.8	-19.2	96.4	-23.0	97.9	-27.8	99.9
			WB	-9.1	82.4	7.0	1.3	-14.1	83.3	-18.2	84.1	-22.1	84.9	-27.1	85.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	25.2	27.9	41.1	52.2	63.1	71.2	72.9	71.1	66.1	53.6	41.1	30.5	(6)
(7)		DBStd	19.39	12.46	12.02	11.34	9.22	8.71	5.82	5.40	5.23	7.27	9.46	9.87	10.54	(7)
(8)		HDD50	2783	771	623	325	83	7	0	0	0	1	69	296	608	(8)
(9)		HDD65	5813	1234	1039	744	390	144	15	5	11	72	373	716	1070	(9)
(10)		CDD50	3320	2	4	48	150	415	637	709	655	485	181	30	4	(10)
(11)		CDD65	874	0	0	2	7	86	202	248	202	107	20	0	0	(11)
(12)		CDH74	8214	0	0	22	133	823	1844	2296	1802	1080	212	2	0	(12)
(13)		CDH80	2595	0	0	1	16	228	598	785	548	373	46	0	0	(13)
(14)	Wind	WSAvg	6.6	8.9	8.7	8.6	8.8	6.8	4.9	3.9	3.6	4.3	6.3	7.3	8.1	(14)
(15)	Precipitation	PrecAvg	38.0	1.9	1.9	2.6	4.1	4.0	4.4	4.4	3.1	3.3	2.8	2.7	2.6	(15)
(16)		PrecMax	49.8	6.7	4.6	6.6	8.3	5.9	8.5	8.0	5.9	7.3	7.5	5.0	6.9	(16)
(17)		PrecMin	31.8	0.2	0.4	0.9	1.8	1.5	1.5	2.3	0.4	0.7	0.6	0.6	1.2	(17)
(18)		PrecStd	4.8	1.5	1.1	1.3	1.8	1.4	1.9	1.5	1.5	1.9	1.7	1.4	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.7	63.8	77.3	82.3	89.0	91.7	94.3	91.8	92.3	84.9	71.3	61.6	(19)
(20)			MCWB	56.2	56.1	63.2	63.3	71.1	75.2	79.4	76.0	73.2	69.8	59.6	57.8	(20)
(21)		2%	DB	53.1	56.9	70.1	78.0	85.6	89.0	89.9	88.6	88.3	80.1	65.7	55.0	(21)
(22)			MCWB	50.3	53.2	60.2	62.2	70.2	74.8	78.0	75.2	72.9	66.0	56.8	52.3	(22)
(23)		5%	DB	47.6	50.4	63.9	74.0	82.5	86.6	87.3	85.7	84.6	75.3	61.3	49.7	(23)
(24)			MCWB	44.7	46.4	55.8	60.8	67.9	73.7	76.9	74.8	71.7	62.5	56.1	46.8	(24)
(25)		10%	DB	41.4	44.2	59.0	69.2	78.9	83.6	84.5	83.1	80.4	70.1	56.9	44.8	(25)
(26)			MCWB	39.3	40.9	53.3	57.7	66.8	71.8	74.8	72.7	68.3	62.0	51.0	42.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.2	59.3	65.2	67.3	74.3	79.8	82.0	80.4	77.1	72.7	63.3	58.8	(27)
(28)			MCDWB	57.6	63.5	76.7	75.8	84.7	88.4	89.9	88.1	87.3	82.0	68.0	61.3	(28)
(29)		2%	WB	50.1	54.9	60.6	64.3	72.5	77.5	80.1	78.0	75.1	69.1	59.3	53.9	(29)
(30)			MCDWB	51.8	57.9	68.4	72.7	82.9	85.8	88.5	85.8	85.1	75.2	63.5	56.3	(30)
(31)		5%	WB	44.3	49.4	57.2	62.1	70.6	75.5	78.3	76.1	73.1	66.1	55.1	47.8	(31)
(32)			MCDWB	46.7	53.2	62.2	69.6	80.0	83.5	86.2	83.0	82.1	70.8	60.1	50.0	(32)
(33)		10%	WB	38.6	42.8	52.9	59.6	68.7	73.9	76.8	74.2	70.8	62.7	50.6	43.6	(33)
(34)			MCDWB	40.8	46.5	57.3	66.9	77.2	82.0	83.9	80.7	77.8	69.6	55.2	45.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.6	16.7	19.1	22.3	20.8	20.4	19.8	21.2	23.4	22.4	18.5	14.1	(35)
(36)			MCDBR	20.9	23.3	25.8	28.2	24.6	23.0	21.6	23.4	26.6	27.6	24.3	19.7	(36)
(37)			MCWBR	18.0	18.2	16.4	15.9	11.5	11.6	11.3	12.5	11.6	14.4	17.5	16.6	(37)
(38)		5% WB	MCDWB	20.4	22.6	23.7	23.2	21.2	19.1	19.8	19.3	21.9	20.9	22.4	18.3	(38)
(39)			MCWBR	18.4	18.6	16.5	15.9	11.5	11.0	11.4	11.5	10.9	13.2	17.6	16.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.316	0.354	0.395	0.436	0.441	0.460	0.457	0.415	0.362	0.315	0.298	(40)	
(41)		taud	2.435	2.386	2.364	2.254	2.192	2.223	2.223	2.151	2.148	2.245	2.393	2.483	2.459	(41)
(42)		Ebn at Noon	274	285	285	279	268	266	260	257	261	265	267	266	(42)	
(43)		Edh at Noon	25	31	35	42	46	44	47	46	39	30	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	568	845	1158	1510	1724	1979	1953	1737	1466	1008	671	479	(44)	
(45)		RadStd	65	84	93	108	146	127	108	109	108	107	64	59	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page